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SYSTEMATIC REVIEW

Accuracy and precision of intraoral scanners for shade matching: A systematic review

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ABSTRACT

Statement of problem. The use of intraoral scanners is rising in prosthetic dentistry; however, systematic analysis of their accuracy and precision for shade matching is scarce.

Purpose. The purpose of this systematic review was to evaluate the accuracy and precision of intraoral scanners for shade matching.

Material and methods. In addition to a manual search, an electronic systematic search was conducted on MEDLINE/PubMed, EMBASE, Web of Science, Cochrane, and Scopus databases. English-language original studies published between January 1, 2010 and March 1, 2022 with intraoral or digital scanners were chosen based on the keywords of tooth color or shade selection or determination, color or shade matching, accuracy, validity, or trueness, and precision, repeatability, or reproducibility as inclusive criteria. Two reviewers independently performed the literature search, selected the studies, collected the data from the studies included, and evaluated the quality of the studies included using a quality assessment method and the Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental Studies. A third reviewer resolved disagreements.

Results. A total of 17 articles concerning the shade matching accuracy and precision of intraoral scanners were selected and reviewed. Among them, 4 articles evaluated only accuracy, 4 articles assessed only precision, and 9 articles investigated both accuracy and precision. Ten articles reported low levels of shade matching accuracy for intraoral scanners, while 11 articles reported high levels of shade matching precision for intraoral scanners.

Conclusions. Based on the current literature, intraoral scanners show acceptable precision but unacceptable accuracy for shade matching. (J Prosthet Dent 2024;132:714-25)

Intraoral scanners are used in a diverse range of applications in various fields of dentistry such as diagnosis of caries,[1-3] monitoring of tooth wear,[4-6] digital smile planning,[7] digital recording for prostheses,[8-18] assessing marginal fit,[19] virtual registration of interocclusal relations,[20-24] obtaining digital diagnostic casts,[25,26] analysis of malocclusions,[27] bracket positioning,[28] imaging in orthodontics,[29,30] dental education,[31] surveying soft tissues,[32] guided implant surgery,[33] aligner fabrication,[34] maxillofacial rehabilitation,[35] diagnosis of periodontal defects,[36] and tooth shade matching.[37,38] Recently, some manufacturers have equipped intraoral scanners with color matching software programs to provide scanners with shade determination ability.[37]

Color measuring devices enhance the objectivity of shade matching as compared with visual shade assessments.[37-40] For optimal performance, such devices should offer straightforward operation, a suitable light source, acceptable optical resolution, satisfactory accuracy, and high precision.[37,41] The performance of dental shade matching devices such as colorimeters, spectrophotometers, digital single-lens reflex and intraoral cameras, and smartphones has been studied.[42-50] Accordingly, spectrophotometers have been reported to have the highest accuracy and precision.[37,41,51]

Accuracy indicates how closely a device measures what is being measured.[37,51] The accuracy of a color

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Clinical Implications

Intraoral scanners currently available are not recommended for tooth shade selection because of their deficient shade matching accuracy in spite of their acceptable shade matching precision.

matching device is assessed by calculating the average differences between the actual quantitative color values (CIELab: L*: lightness, a*: redness-greenness, b*: yellowness-blueness; CIELCh: L*: lightness, C: chroma, h: hue angle) of a set of specimens and the device-recorded quantitative color values for the same specimens and/or by calculating the percentage of accurate color determination recorded by the device for the specimens regarding quantitative color values or qualitative/nominal color data.[43,51,52]

Precision is the degree to which a measuring device provides consistent results.[37,51] The precision of a color matching device is measured by calculating the average differences between the quantitative color results of the device in repeated measurements for similar specimens under similar conditions and/or by calculating the percentage of repeatable or reproducible color determination recorded by the device for the same specimens considering quantitative color values or qualitative color data.[51,53,54] Consequently, the accuracy and precision are estimated based on percentages and/or $\Delta E_{ab}/\Delta E_{00}$ color difference units calculated from formulas such as $\Delta E_{ab}=\sqrt{(\Delta L)^2+(\Delta a)^2+(\Delta b)^2}$ and $\Delta E_{00} = \sqrt{(\frac{\Delta L'}{K_L S_L})^2+(\frac{\Delta C'}{K_C S_C})^2+(\frac{\Delta H'}{K_H S_H})^2+R_T\times(\frac{\Delta C'}{K_C S_C})^2\times(\frac{\Delta H'}{K_H S_H})}$.[37] As the calculated color difference decreases or the calculated percentage increases, the device accuracy and precision increase. To clinically evaluate the accuracy and precision, the color difference is compared with visual thresholds.[55-59] A 50:50% perceptibility threshold refers to a situation in which 50% of observers detect a color difference between 2 objects while the other 50% detect no color difference.[56-59] A 50:50% acceptability threshold refers to a situation where a color difference is acceptable for 50% of observers.[56-59] A perceptibility threshold of ΔE_{ab}=1.2 and ΔE_{00}=0.8 and an acceptability threshold of ΔE_{ab}=2.7 and ΔE_{00}=1.8 have been established (ISO/TR 28642:2016).[56,59]

Current intraoral scanners with color matching tools can only assess qualitative color data (Fig. 1). They have been marketed for tooth shade selection by the manufacturers; however, their shade matching performance has been questioned,[37] and the shade matching accuracy and precision of intraoral scanners are unclear. Thus, the purpose of this systematic review was to assess the literature for the accuracy and precision of intraoral scanners for shade matching. The research hypothesis was that intraoral scanners would indicate acceptable shade matching accuracy and precision compared with a standard device or method in tooth shade selection.

MATERIAL AND METHODS

This systematic review conformed to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[60] A problem, intervention, comparison, outcome (PICO) strategy was used to formulate the research question: "In dental shade selection, do intraoral scanners, compared with a standard device or method, show acceptable shade matching accuracy and precision?". The PICO strategy was also applied to formulate the literature search (Table 1). An electronic systematic search was conducted on MEDLINE/PubMed, EMBASE, Web of Science, Cochrane, and Scopus databases in addition to a manual search. All search terms were MeSH terms.

English-language studies published between January 1, 2010 and March 1, 2022 with an abstract and full text were initially included. The titles and abstracts were assessed to exclude review articles of intraoral scanners, articles related to the use of intraoral scanners in disciplines such as orthodontics, periodontics, and maxillofacial surgery, and articles related to caries diagnosis, tooth wear monitoring, digital scanning for prostheses, marginal fit evaluation, virtual registration of interocclusal relations, dental education, soft tissue examination, implant surgery, aligner fabrication, maxillofacial rehabilitation, and diagnosis of periodontal diseases. All titles and abstracts were evaluated to include in vitro or clinical studies that assessed the ability of intraoral scanners for tooth shade matching, tooth color determination, or dental shade selection. In addition, the reference lists of the included articles were searched to add relevant articles. Nonpeer-reviewed literature, theses, and unpublished materials were excluded. The process of literature search and article selection was performed independently by 2 calibrated reviewers (F.T., M.N.).

The same reviewers extracted the data from the articles to designed tables in which the name of the intraoral scanner, the manufacturer, the sample size, the type of specimens, the type of experiment, the specimen's measured region, the level of accuracy of the intraoral scanner, the level of precision of the intraoral scanner, and the control device or method were collected. Disagreements were resolved by a third calibrated reviewer (M.M.). A quality assessment was done by the same 2 reviewers individually to determine a quality score (QS) for the included studies based on 8 criteria: the type of

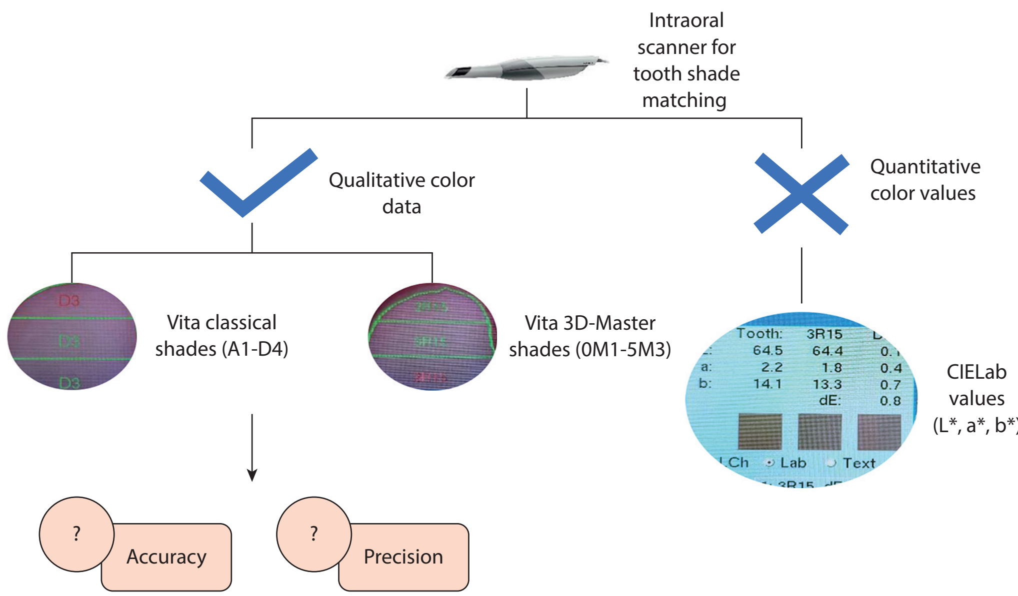

Figure 1. Flow diagram for tooth shade matching of intraoral scanners.

Table 1. Literature search strategy according to PICO elements

PICO Elements	Keywords	Search Terms	Search Strategy
P: problem/patient/ population/participant	Tooth shade selection	Tooth shade selection, tooth color determination, dental shade selection	tooth* shade* OR shade* selection* OR shade* determination* OR tooth* color* OR color* selection* OR color * determination* OR tooth* shade* selection* OR dental* shade* determination* OR dental* shade* selection*
I: intervention	Intraoral scanner	Intraoral scanner, digital scanner	intraoral* OR digital* OR scanner*
C: comparison	Colorimeter, spectrophotometer, spectroradiometer	Colorimeter, spectrophotometer, spectroradiometer	colorimeter* OR spectrophotometer* OR spectroradiometer*
O: outcome	Shade matching accuracy and precision of intraoral scanner	Accuracy, precision, validity, trueness, reliability, reproducibility, performance	accuracy* OR precision* OR validity* OR trueness* OR reliability* OR reproducibility* OR performance*

specimen; the type of experiment; the specimen region measured; information about the devices, operators, or operations; accuracy; precision; appropriate statistical methods; and the lucidity of conclusions (Table 2).[61] They also independently assessed the quality of the articles based on the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental Studies (Table 3),[62] and the third reviewer resolved any disagreements. The individual and overall risk of bias for the included articles were determined based on the answers (yes: low risk; no: high risk; unclear: unclear risk) to the questions of the JBI Checklist by calculating the percentage of the answer type (yes, no, unclear) for each question considering all included articles. Also, the included articles were categorized into in vivo or in vitro studies, and the results on the shade matching accuracy and precision of intraoral scanners were then assessed regarding in vivo and in vitro studies separately and

Table 2. Quality assessment and quality score determination based on 8 criteria

Criteria	Type	Points
Specimens	Natural teeth/shade tabs	1
	Extracted teeth	0
Experiment	In vivo/model	1
	In vitro	0
Specimen's measured region	Entire tooth/3 regions	1
	Middle/central region	0
Information about devices/ operators/operations	Clear	1
	Not clear	0
Is accuracy reported?	Yes	1
	No	0
Is precision reported?	Yes	1
	No	0
Appropriate statistical methods	Yes	1
	No	0
Conclusions	Clear	1
	Not clear	0

Table 3. Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental Studies (nonrandomized experimental studies)

Criteria	Yes	No	Unclear	Not Applicable
1. Q1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	☐	☐	☐	☐
2. Q2. Were the participants included in any comparisons similar?	☐	☐	☐	☐
3. Q3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	☐	☐	☐	☐
4. Q4. Was there a control group?	☐	☐	☐	☐
5. Q5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	☐	☐	☐	☐
6. Q6. Was follow-up complete and if not, were differences between groups in terms of their follow-up adequately described and analyzed?	☐	☐	☐	☐
7. Q7. Were the outcomes of participants included in any comparisons measured in the same way?	☐	☐	☐	☐
8. Q8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Q9. Was appropriate statistical analysis used?	☐	☐	☐	☐

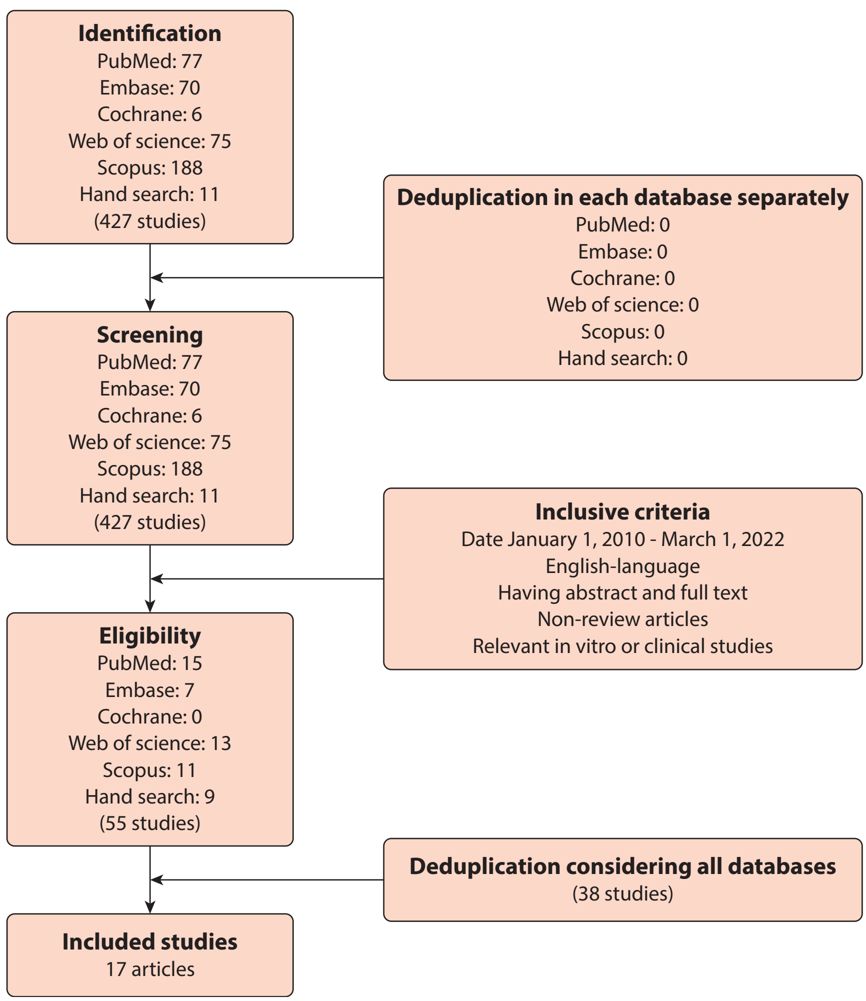

Figure 2. PRISMA flow diagram for processes of literature search and article selection. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

jointly by the 2 reviewers. The third reviewer resolved any disagreements.

RESULTS

The literature search and article selection processes are presented in Figure 2. A total of 427 articles were found after electronic and manual searches. No database search showed any duplicated articles. After applying the inclusion criteria, 55 articles were identified. Among them, 38 articles were duplicates and were excluded. Consequently, 17 articles[63-79] published between January 1, 2017 and March 1, 2022 were included in this review

Table 4. Collected data and quality scores of articles included

Reference	Intraoral Scanner	Manufacturer	Color Measurement	Sample Size	Type of Specimens	Type of Experiment	Measured Area	Quality Score (Max=8)	Accuracy	Precision
Brandt et al,[63] 2017	TRIOS Color	3Shape A/S	Qualitative (based on Vita 3D-Master), Quantitative (based on CIELab and CIELCh values obtained from custom conversion chart)	107 (107 participants, 1 tooth each)	Natural teeth	In vivo	Middle/central region	5	43.9% (with reference to Vita Easyshade Advance 4.0 spectrophotometer), 78.5% (with reference to acceptability threshold of ΔE_{ab}=6.8)	78.3% (comparable with Vita Easyshade Advance 4.0); good to excellent
Mehl et al,[66] 2017	TRIOS Color	3Shape A/S	Qualitative (based on Vita 3D-Master), Quantitative (based on CIELab values obtained from custom conversion chart)	40 (20 participants, 2 teeth each)	Natural teeth	In vivo	Middle/central region	6	61.2% (with reference to major shade chosen by 7 color measuring methods)	66.7%; comparable to Vita Easyshade, SpectroShade, and SpectroShade Micro spectrophotometers
Yoon et al,[64] 2018	TRIOS Pod Color-P13	3Shape A/S	Quantitative (based on CIELab values: screen shots made from scanned images and read with ADOBE Photoshop CS6)	5	Shade tabs (A1, A2, A3, A3.5, A4)	In vitro	Middle/central region	6	ΔE_{ab} =7.0 ±1.2 (A1), 13.6 ±0.8 (A2), 12.1 ±0.5 (A3), 11.6 ±0.7 (A3.5), and 10.5 ±0.5 (A4) (with reference to ShadeEye-NCC colorimeter)	99.7% for L* value, 94.5% for a* value, and 99.3% for b* value; excellent
Culic et al,[72] 2018	CEREC Omnicam with CEREC SW 4.5.2 software	Dentsply Sirona	Qualitative (based on Vita classical and Vita 3D-Master)	80 (4 participants, 20 teeth each)	Natural teeth	In vivo	Entire tooth/3 tooth regions	5	15% (17.5% based on Vita classical and 12.9% based on Vita 3D-Master with reference to Easyshade Advance 4.0)	Not reported
Liberato et al,[65] 2019	TRIOS	3Shape A/S	Qualitative (based on Vita classical and Vita 3D-Master)	28 (28 participants, 1 tooth each)	Natural teeth	In vivo	Middle/central region	6	Not reported	Fleiss kappa of .639 based on Vita classical and Fleiss kappa of .874 based on Vita 3D-Master; good to excellent
Reyes et al,[67] 2019	TRIOS	3Shape A/S	Qualitative (based on Vita 3D-Master)	10 (10 participants, 1 tooth each)	Natural teeth	In vivo	Middle/central region	6	Not reported	86.67% ±17.21% for value, 90.00% ±16.10% for hue, and 81.11% ±16.60% for chroma; 86.66% ±11.48% (overall)
Liu et al,[68] 2019	TRIOS 3	3Shape A/S	Quantitative (based on CIELab values: RGB values of scanned images were converted to CIELab values of resin blocks printed with a 3D printer using training samples and a developed color profile)	18 (18 resin blocks)	Ceramage, Shofu resin blocks in 16 body shades (A1 to D4) and 2 gum shades (GUM-L, GUM-D)	In vitro	Middle/central region	4	ΔE_{ab}=6.5378 (ranging from 2.1850 to 11.2156); clinically unacceptable (ΔE_{ab}>6.80) with reference to ColorEye 7000A spectrophotometer	Not reported
Yılmaz et al,[70] 2019	TRIOS	3Shape A/S	Qualitative (based on Vita 3D-Master)	5 (5 participants. 1 tooth each)	Natural teeth	In vivo	Entire tooth/3 tooth regions	6	Acceptable (comparable to Vita Easyshade Compact spectrophotometer)	Not reported
Revilla-León et al,[69] 2020	TRIOS 3	3Shape A/S	Qualitative (based on Vita classical and Vita 3D-Master)	3 (1 participant, 3 teeth each)	Natural teeth	In vivo	Middle/central region	6	Not reported	Unacceptable (compared with Vita Easyshade V spectrophotometer and under different lighting conditions)
Rutkūnas et al,[71] 2020	TRIOS 3	3Shape A/S	Qualitative (based on Vita classical and Vita 3D-Master), Quantitative (based on CIELab values obtained from a database of SpectroShade spectrophotometer)	120 (20 participants, 6 teeth each)	Natural teeth	In vivo	Middle/central region	7	53.3% based on Vita 3D-Master and 27.5% based on Vita classical; ΔE_{ab} <3.7 for 51.7% of cases based on Vita 3D-Master and for 21.7% of cases based on Vita classical (with reference to SpectroShade spectrophotometer)	90.3% based on Vita 3D-Master and 87.2% based on Vita classical; good interoperator precision (Cohen kappa of >0.75)

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Table 4. (*Continued*) Collected data and quality scores of articles included

Reference	Intraoral Scanner	Manufacturer	Color Measurement	Sample Size	Type of Specimens	Type of Experiment	Measured Area	Quality Score (Max=8)	Accuracy	Precision
Hampé-Kautz et al,[74] 2020	1) TRIOS 3; 2) CEREC Omnicam	1) 3Shape A/S; 2) Dentsply Sirona	Qualitative (based on Vita 3D-Master), Quantitative (based on CIELab values obtained from a conversion chart)	40 (40 participants, 1 tooth each)	Natural teeth	In vivo	Middle/central region	4	ΔE_{00}=3.1 and L*=83.42 ±0.16 for TRIOS 3; ΔE_{00}=2.75 and L*=82.56 ±0.34 for CEREC Omnicam; unacceptable with reference to Vita Easyshade V spectrophotometer (L*=78.06 ±0.50) and threshold of ΔE_{00}=2.45	Not reported
Ebeid et al,[73] 2021	1) CEREC Omnicam with CEREC software version 4.6; 2) CEREC Primescan with CEREC software version 5.5.1; 3) TRIOS with 3-cart version	1,2) Dentsply Sirona; 3) 3Shape A/S	Qualitative (based on Vita classical)	10 (10 disk-shaped ceramic specimens,10 mm in diameter and 1 mm thick)	Vitablocs Mark II monochromatic disks (A1, A2, A3, A3.5, A4, B2, B3, C2, C3, and D3)	In vitro	Middle/central region	6	57% for CEREC Omnicam, 63% for CEREC Primescan, and 66% for TRIOS (with reference to Vitablocs Mark II Vita classical shades)	51.9% for CEREC Omnicam, 48.4% for CEREC Primescan, and 51.7% for TRIOS
Fattouh et al,[76] 2021	TRIOS 3	3Shape A/S	Qualitative (based on Vita 3D-Master), Quantitative (based on CIELab values obtained from a conversion chart)	50 (50 participants, 1 tooth each)	Natural teeth	In vivo	Middle/central region	6	Not reported	94% ±4.6% (comparable with Vita Easyshade Advance with the precision of 93% ±4.02%)
Czigola et al,[75] 2021	TRIOS 3	3Shape A/S	Qualitative (based on Vita classical and Vita 3D-Master), Quantitative (based on CIELab values obtained from a conversion chart)	30 (10 participants, 3 teeth each)	Natural teeth	In vivo	Entire tooth/3 tooth regions for central teeth and Middle/central region for premolar and molar teeth	8	21.64% (with reference to the best matched shade chosen by examiners, a supervisor, and participants); ΔE_{00}=4.29	Intraoperator precision of 100% (regarding cervical region of central tooth) and 56% (regarding central, premolar, and molar teeth) based on Vita 3D-Master shades; ΔE_{00}=0 (regarding central and premolar teeth), ΔE_{00}=1.09 (regarding central, premolar, and molar teeth)
Sirintawat et al,[77] 2021	TRIOS 3 Basic	3Shape A/S	Qualitative (based on Vita 3D-Master), Quantitative (based on CIELab values obtained from a conversion chart)	30 (30 milled ceramic crowns)	Vitablocs Mark II monochromatic ceramic crowns (1M1, 1M2, 2M1, 2M2, 2M3, 3M1, 3M2, 3M3, and 4M2)	In vitro	Middle/central region	6	ΔE_{ab}=5.96 ±1.83 (with reference to Vitablocs Mark II 3D-Master shades)	Intraclass correlation coefficients of 0.996 for L*, 0.994 for a*, and 0.884 for b*
Ebeid et al,[78] 2022	1) CEREC Omnicam with CEREC software version 4.6; 2) CEREC Primescan with CEREC software version 5.5.1; 3) TRIOS 3; 4) TRIOS 4	1, 2) Dentsply Sirona; 3,4) 3Shape A/S	Qualitative (based on Vita classical)	20 (20 participants, 1 tooth each)	Natural teeth	In vivo	Middle/central region	7	55% for CEREC Omnicam, 69% for CEREC Primescan, 75% for TRIOS 3, and 76% for TRIOS 4 (with reference to most frequent visual shade selected by 10 observers for each tooth)	77% for CEREC Omnicam, 82% for CEREC Primescan, 79% for TRIOS 3, and 82% for TRIOS 4
Huang et al,[79] 2022	1) TRIOS 3; 2) TRIOS 4	3Shape A/S	Qualitative (based on Vita classical and Vita 3D-Master)	130 (23 participants, totally130 teeth)	Natural teeth	In vivo	Entire tooth/3 tooth regions	8	TRIOS 3: 43-63% based on Vita classical and 27-28% based on Vita 3D-Master; TRIOS 4: 6-40% based on Vita classical and 3% based on Vita 3D-Master (with reference to Vita Easyshade V)	TRIOS 3: 75% based on Vita classical and 76% based on Vita 3D-Master; TRIOS 4: 72% based on Vita classical and 64% based on Vita 3D-Master

(Fig. 2). No new articles were found from reference lists. Regarding the shade matching performance of intraoral scanners, 4 articles[68,70,72,74] evaluated only accuracy, 4 articles[65,67,69,76] assessed only precision, and 9 articles[63,64,66,71,73,75,77-79] assessed both accuracy and precision. The name of the intraoral scanner, the manufacturer, the sample size, the type of specimen, the type of experiment, the specimen region measured, the level of accuracy and precision, the control device or method, and the QS of the included articles[63-79] are shown in Table 4.

For article selection after assessing the titles and abstracts, significant agreement was found between the 2 reviewers for the articles included (Cohen kappa=0.97, P<.001) and excluded (Cohen kappa=0.97, P<.001). For quality assessment, significant agreement was found between the reviewers in terms of the type of specimen (percent of agreement=100%), the type of experiment (Cohen kappa=1.00, P<.001), the specimen region measured (Cohen kappa=1.00, P<.001), the information about devices, operators, or operations (Cohen kappa=0.72, P<.001), the accuracy reported (Cohen kappa=1.00, P<.001), the precision reported (Cohen kappa=1.00, P<.001), the appropriate statistical methods (Cohen kappa=0.64, P<.001), and the lucidity of conclusions (percent of agreement=82%). Regarding all 8 criteria, significant agreement was found between the reviewers (Cohen kappa=0.88, P<.001). After assessing the articles on the JBI Checklist, significant agreement between the reviewers was found (percentage of agreement=84%, Cohen kappa=0.55, P<.001). The number and percentage of the articles scoring either 0 or 1 of the 8 criteria are shown in Figure 3.

The individual and overall risk of bias for the included articles[63-79] based on the JBI Checklist are presented in Table 5. The percentages of risk of bias for the included articles[63-79] are presented in Figure 4. For all included articles,[63-79] a 100% low risk of bias was found for question 1, while an 88% high risk of bias was found for question 6 (Fig. 4). For questions 2, 3, 4, 5, 7, 8, and 9, a low risk of bias between 59% and 94% was found for the included articles[63-79] (Fig. 4). Overall, the risk of bias for the included studies[64,65,67-73,75-79] was low, except for 1 study[74] with a high risk of bias and 2 studies[63,66] with an unclear risk of bias (Table 5).

The shade matching accuracy for TRIOS 3, TRIOS 4, CEREC Omincam, and CEREC Primescan ranged between 3%[79] and 78.5%,[63] between ΔE_{ab}=6.0[77] and ΔE_{ab}=13.6,[64] and between ΔE_{00}=2.45[74] and ΔE_{00}=4.29[75] (Table 4). Also, their shade matching precision ranged between 48.4%[73] and 100%,[75] between ΔE_{00}=0[75] and ΔE_{00}=1.09,[75] between Fleiss kappa=0.639[65] and Fleiss kappa=0.874,[65] and between ICC=0.884[77] and ICC=0.996[77] (ICC: intraclass correlation coefficient), which was defined as good to excellent[63-65] precision (Table 4).

Among the studies included, 13 studies[63,65-67,69-72,74-76,78,79] were in vivo, evaluating natural teeth, while 4 studies[64,68,73,77] were in vitro, assessing shade tabs, resins, or ceramics. Separating the results of in vivo from in vitro studies did not affect the overall results. Among the included articles, 10 articles[64,66,68,71-75,77,79] reported low shade matching accuracy for intraoral scanners as opposed to 3 articles[63,70,78] that reported otherwise, while 11 articles[63-67,71,75-79] reported high shade matching precision for intraoral scanners in contrast with 2 articles[69,73] that reported low precision (Table 6).

DISCUSSION

The shade matching accuracy and precision of intraoral scanners were separately assessed. Only 3 of the included studies[63,70,78] concluded that shade matching accuracy for intraoral scanners was acceptable. Two of these studies[63,78] reported low to moderate accuracies of 43.9% (with reference to Vita Easyshade Advance 4.0 spectrophotometer)[63] and 78.5% (with reference to an acceptability threshold of ΔE_{ab}=6.8)[63] for TRIOS Color, 55% for CEREC Omnicam, 69% for CEREC Primescan, 75% for TRIOS 3, and 76% for TRIOS 4 (with reference to the most frequent shade chosen by 10 observers).[78] Because of an unclear risk of bias for the study[63] and its high assumed threshold, the reported accuracy[63] might have been overestimated. The reported accuracy[63] seems unacceptable because it exceeded an acceptability threshold of ΔE_{ab}=2.7.[56,59]

Another included study[78] reported an accuracy between 55% and 76% for intraoral scanners in comparison with visual shade assessment using a shade guide (VITA classical A1-D4 shade guide; VITA Zahnfabrik H. Rauter GmbH & Co KG). However, visual assessment using a shade guide is not regarded as a clinical standard in dental shade selection.[37-41,50-53] Only 1 study[70] reported acceptable accuracy for TRIOS similar to that of a spectrophotometer (Vita Easyshade Compact; VITA Zahnfabrik H. Rauter GmbH & Co KG). Though measuring under 2 light sources of 4000 and 6500 K might be an advantage of this study,[70] testing only a few human teeth and shades (n=5) might have been a source of bias.

The low shade matching accuracy of intraoral scanners may stem from the nature of scanning. Regardless of scanning technology, confocal laser scanning, continuous filming and triangulation, or active wave-front sampling, scanning still cannot provide standard illumination and readings for accurate shade selection such as 2×45 degrees, polarized, telecentric, monochromatic illumination and 0 degrees, polarized, telecentric reading as opposed to spectrophotometry.[37] Though TRIOS 3, TRIOS 4,

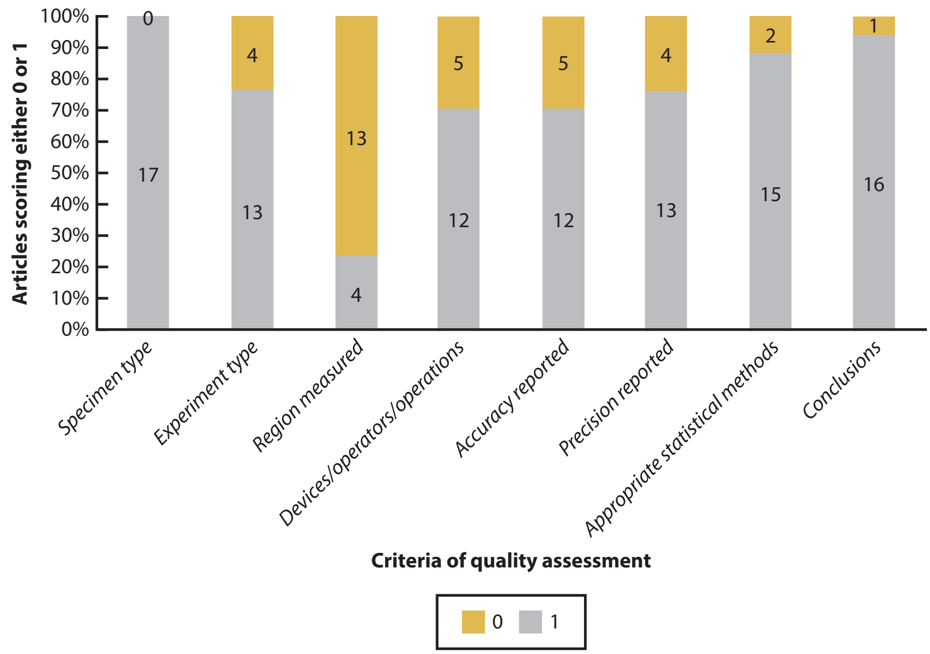

Figure 3. Percentage and number of included articles based on points obtained from 8 criteria of quality assessment of Table 2.

Table 5. Risk of bias for included articles based on Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental studies

	Risk of Bias									
Reference	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Overall**
Brandt et al,[63] 2017	Low	Low	Unclear	Low	High	High	High	High	Low	Unclear
Mehl et al,[66] 2017	Low	High	Unclear	High	Low	High	High	Low	Low	Unclear
Yoon et al,[64] 2018	Low	Low	High	Low	Low	High	Low	Low	Low	Low
Culic et al,[72] 2018	Low	Low	Unclear	Low	High	High	Low	High	Unclear	Low
Liberato et al,[65] 2019	Low	Low	Unclear	Low	Low	High	Low	Unclear	Low	Low
Reyes et al,[67] 2019	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Liu et al,[68] 2019	Low	Low	Unclear	Low	Unclear	High	Low	Unclear	Low	Low
Yılmaz et al,[70] 2019	Low	Low	Low	Low	High	High	Low	Low	Low	Low
Revilla-León et al,[69] 2020	Low	Low	Low	Low	Low	High	Low	Low	Low	Low
Rutkūnas et al,[71] 2020	Low	Low	Low	Low	Low	High	Low	Low	Low	Low
Hampé-Kautz et al,[74] 2020	Low	Low	Unclear	Low	High	High	High	Unclear	High	High
Ebeid et al,[73] 2021	Low	Low	Low	Low	Low	High	Low	Low	Low	Low
Fattouh et al,[76] 2021	Low	Low	Low	Low	Unclear	High	High	Unclear	Low	Low
Czigola et al,[75] 2021	Low	Low	Low	Low	Low	Low	Low	Low	Low	Low
Sirintawat et al,[77] 2021	Low	Low	Low	Low	Low	High	High	Low	Low	Low
Ebeid et al,[78] 2022	Low	Low	Low	Low	Low	High	Low	Low	Low	Low
Huang et al,[79] 2022	Low	Low	Low	Low	Low	High	Low	Low	Low	Low

CEREC Omnicam, and CEREC Primescan are equipped with light-emitting diode light sources similar to those of spectrophotometers,[37] their inaccuracies may be rooted in their unpolarized illumination that records brighter shades for the tooth compared with the actual tooth shade.[74] Also, reading at diverse angles rather than 0 degrees, which inevitably occurs in scanning, may explain inaccuracies.[37] Another reason may be their inability to measure quantitative color values. Scanners use shade matching software programs to find the best matched shade from Vita classical and Vita 3D-Master shade guides (VITA Toothguide 3D-MASTER; VITA Zahnfabrik H. Rauter GmbH & Co KG). The software programs may cause errors. For example, 1 included study[72] showed better results for CEREC Omnicam with Vita classical than with Vita 3D-Master, while another included study[71] reported worse results for TRIOS 3 with Vita classical than with Vita 3D-Master (Table 4).

Most of the included studies[63-67,71,75-79] reported high shade matching precision for intraoral scanners, except for 2 included studies,[69,73] 1 of which tested TRIOS 3 under 4 different ambient lights (10 000, 1000, 500, and

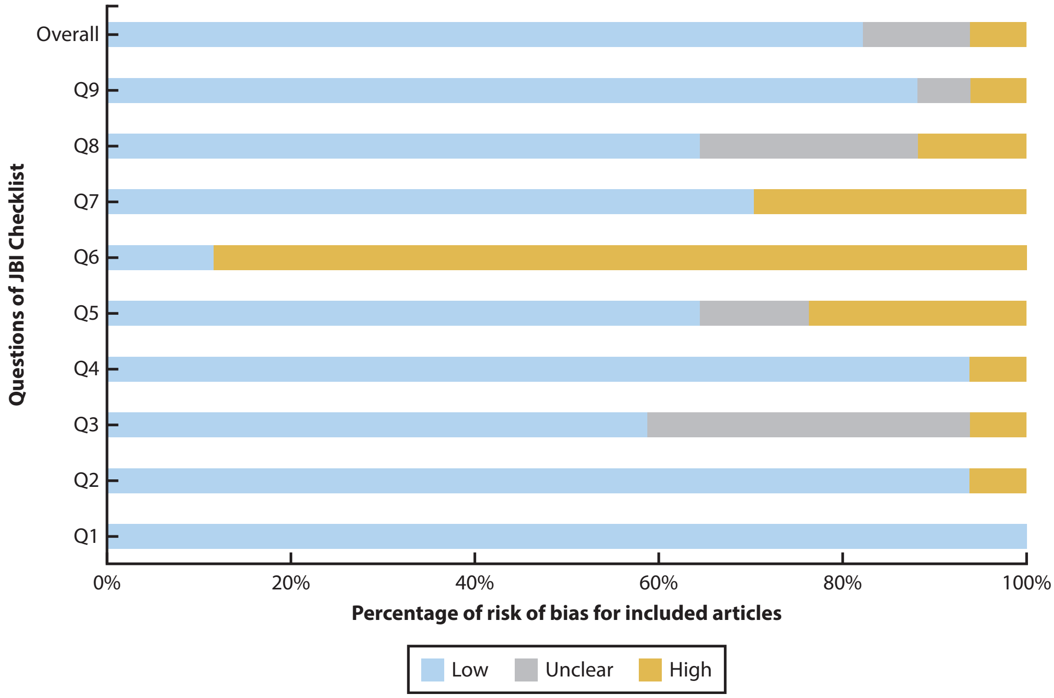

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Figure 4. Percentages of risk of bias for included articles based on questions of Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental studies.

0 lx).[69] As ambient light affects visual and instrumental shade selection, most of the included studies[63-68,71-79] provided stable light conditions to reduce the risk of bias. In another included study,[73] color measuring was conducted on a small central area of ceramic disks under an ambient light of 6500 K, but determining the same area of a disk on the images captured by repeated scanning could be imprecise. This may be why low precision was reported by this study[73] (Table 4). Considering a perceptibility threshold of ΔE_{00}=0.8 and an acceptability threshold of ΔE_{00}=1.8,[56,59] the reported precisions[63-67,71,75-79] seem acceptable. This may be linked to the scanners' built-in high precision and consistent conditions of the experiments.

The included studies reported a wide range of shade matching accuracy and precision for intraoral scanners. The wide range stems from diverse factors affecting shade results such as ambient light,[67,69] software program,[64,71,72,78,79] tooth type (central, premolar, molar),[75] tooth region (cervical, body, incisal),[70,72,75,79] type of color measurement (qualitative, quantitative),[63,66,72] operator experience,[65,75] calibration of scanning,[63,67,73,75] threshold,[63,68,74] scanner brand and model,[73,74] conversion chart,[63,66,71,74-77] specimen type,[68,73,77] experiment type (in vivo or in vitro),[61] control group or device for accuracy assessment,[63,66,70,72,74,75,78] and type of precision assessment (intraoperator or device, interoperator or device, time, or specimen manipulation).[37,51,63,66,67,77]

Table 6. Results on levels of shade matching accuracy and precision of intraoral scanners reported by included studies based on types of experiment and specimen

Reference	Experiment Type	Specimen Type	Accuracy	Precision
Brandt et al[63]	In vivo	Natural teeth	−/+	+
Mehl et al[66]	In vivo	Natural teeth	−	+
Culic et al[72]	In vivo	Natural teeth	−	Not reported
Liberato et al[65]	In vivo	Natural teeth	Not reported	+
Reyes et al[67]	In vivo	Natural teeth	Not reported	+
Yilmaz et al[70]	In vivo	Natural teeth	+	Not reported
Revilla-Leon et al[69]	In vivo	Natural teeth	Not reported	−
Rutkūnas et al[71]	In vivo	Natural teeth	−	+
Hampé-Kautz et al[74]	In vivo	Natural teeth	−	Not reported
Fattouh et al[76]	In vivo	Natural teeth	Not reported	+
Czigola et al[75]	In vivo	Natural teeth	−	+
Ebeid et al[78]	In vivo	Natural teeth	−/+	+
Huang et al[79]	In vivo	Natural teeth	−	+
Yoon et al[64]	In vitro	Shade tab	−	+
Liu et al[68]	In vitro	Resin	−	Not reported
Ebeid et al[73]	In vitro	Ceramic	−	−
Sirintawat et al[77]	In vitro	Ceramic	−	+

−: unacceptable +: acceptable −/+: at the border of unacceptability and acceptability.

A majority of the included studies[63,65-67,69-72,74-76,78,79] were conducted in vivo which can be advantageous because evaluation of natural teeth simulates clinical settings better than in vitro studies[64,68,73,77] (Table 6). Although shade measuring conditions may vary for natural teeth as compared with shade tabs, resins, or ceramics, excluding the results of in vitro studies from the review did not alter the overall results obtained from both in vivo and in vitro studies (Table 6). Also, as tooth shade reproduction consists of both in vivo (clinical) and in vitro (laboratory) processes, the results in the present review were reported considering both types of studies. Because the tooth shade varies in different tooth regions, the shade should be assessed in 3 regions (cervical, body, incisal), thus recording a 3-shade report.[52,54] However, only 4 included studies[70,72,75,79] recorded 3-shade reports. A control group, device, or method plays a crucial role in determining the accuracy of shade matching devices. The control was a spectrophotometer (VITA Easyshade Advance 4.0 or Compact; VITA Zahnfabrik H. Rauter GmbH & Co KG) for 3 included studies,[63,70,72] a spectrophotometer (VITA Easyshade V; VITA Zahnfabrik H. Rauter GmbH & Co KG) for 2 included studies,[74,79] a benchtop spectrophotometer (ColorEye 7000A; X-Rite Inc) for 1 included study,[68] a spectrophotometer (SpectroShade Micro; MHT Optic Research AG) for 1 included study,[71] a colorimeter (ShadeEye-NCC; Shofu Inc) for 1 included study,[64] a major shade chosen by all methods tested for 1 included study,[66] the best matched shade chosen by different observers for 2 included studies,[75,78] and the shade of monochromatic ceramics (VITABLOCS Mark II; VITA Zahnfabrik H. Rauter GmbH & Co KG) for 2 included studies.[73,77] Spectrophotometers and spectroradiometers, as the most accurate devices for color measurement,[37,41,51,56] seem appropriate as a control. However, the use of diverse controls by the included studies has made the comparison of their results difficult. Also, factors such as device calibration, device shade guide settings, operator calibration, ambient light, and specimen color type (monochromatic ceramics versus polychromatic teeth) might have affected the results.

Six included studies[63,66,74-77] used conversion charts to convert the qualitative color data (the shades of Vita classical and Vita 3D-Master) recorded with intraoral scanners to CIELab and/or CIELCh values and to measure $\Delta E_{ab}/\Delta E_{00}$ color differences between the records of scanners and the control. One of these studies[63] customized a conversion chart by using the Vita 3D-Master shade tabs recorded by a spectrophotometer (Vita Easyshade; VITA Zahnfabrik H. Rauter GmbH & Co KG), while another study[66] customized a conversion chart using the Vita 3D-Master shade tabs recorded by a SpectroShade Micro spectrophotometer. The other evaluated studies[74-77] estimated CIELab values using different premade conversion charts. The use of conversion charts may cause errors, confounding the shade results. Because of the considerable differences among these studies and especially the scales of measurement used, it was not possible to calculate an average accuracy or precision score for a meta-analysis.

Based on this review, current intraoral scanners with shade matching ability are precise but not accurate for dental shade selection; as a result, they cannot be considered as a standard device for shade prediction. Hence, these devices are not clinically implicated when determining a shade for a tooth or a restoration. Given that only 2 included articles[75,79] gained the QS of 8 out of 8 (Table 4) and only 2 included articles[67,75] met all the criteria (Tables 3 and 5), future studies with rigid methodologies are needed to verify the performance of intraoral scanners. As current intraoral scanners still do not meet the requirements for accurate tooth shade selection, aiming at technologically enhancing their shade matching accuracy is suggested for future research.

CONCLUSIONS

Based on the findings of the present systematic review, the following conclusions were drawn:

1. Intraoral scanners show unacceptable shade matching accuracy.
2. Intraoral scanners indicate acceptable shade matching precision.

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